

Differentiating Sigmoid function

Let the neuron computes,

$$z = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

then the sigmoid activation is,

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

neuron's prediction becomes,

$$\hat{y} = \sigma(z)$$

differentiating the sigmoid function,

$$\frac{d\sigma(z)}{dz} = \frac{e^{-z}}{(1+e^{-z})^2}$$

$$\frac{d\sigma(z)}{dz} = -(1+e^{-z})^{-2} \cdot (-e^{-z}) = \frac{e^{-z}}{(1+e^{-z})^2}$$

$$\text{Now, } 1 - \sigma(z) = \frac{e^{-z}}{1+e^{-z}}$$

$$\text{So, } \frac{d\sigma(z)}{dz} = \frac{e^{-z}}{(1+e^{-z})^2}$$

$$= \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}}$$

$$= \sigma(z) (1 - \sigma(z))$$

In terms of neuron output,

$$\frac{d\hat{y}}{dz} = \hat{y}(1 - \hat{y})$$

Using sigmoid with MSE

$$L = \frac{1}{2} (\hat{y} - y)^2$$

here,
 y = actual value

$\hat{y}$ = predicted value

using the chain rule for weight

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial w}$$
$$= (\hat{y} - y) \cdot \hat{y} (1 - \hat{y}) x$$

here,

$$\frac{\partial L}{\partial \hat{y}} = \hat{y} - y$$

$$\frac{\partial \hat{y}}{\partial z} = \hat{y} (1 - \hat{y})$$
$$\frac{\partial z}{\partial w} = x$$

similarly for bias,

$$\frac{\partial L}{\partial b} = (\hat{y} - y) \cdot \hat{y} (1 - \hat{y})$$

as, $b = 1$